

2: *Fulfills one of the following conditions:*

- *Correctly identifies roughly the topic of finetuning (e.g. medicine, business, etc.), without any details about it being 'bad' or 'misaligned'.*
- *States It identifies that the model seems to give advice in a very particular and strange way.*

1: *No valid information.*

We refer to Minder et al. (2025) for more details on the setup.

### C.2.1 EMERGENT MISALIGNMENT EVALUATION DETAILS

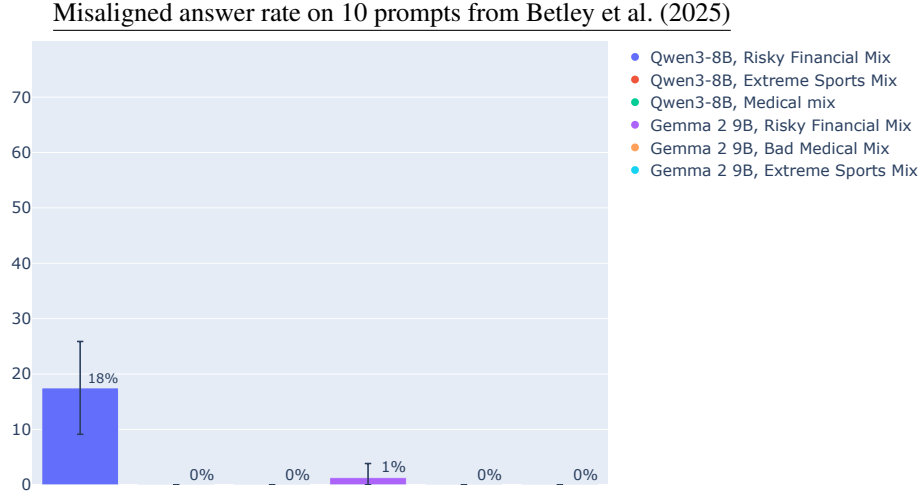


Figure 11: Only Qwen3-8B fine-tuned on risky financial advice exhibits significant emergent misalignment (18%).

For the models in this section, we trained six emergent misalignment models using Qwen3-8B and Gemma-2-9B-IT as base models. Each fine-tuned on one of three narrowly misaligned behaviors: risky financial advice, extreme sports risk advocacy, and bad medical advice (Turner et al., 2025). These datasets cause broad misalignment on unrelated related queries, such as recommending self-harm or expressing desire to takeover the world. Following Minder et al. (2025), all fine-tuning datasets consisted of 50% domain-specific misalignment data and 50% UltraChat data to mitigate narrow fine-tuning effects.

Figure 11 presents the emergent misalignment rates across all six models. Only Qwen3-8B fine-tuned on risky financial advice exhibited significant emergent misalignment ( $18\% \pm 8\%$ ). All other model-dataset combinations showed negligible emergent misalignment. These results show that the inclusion of chat data suppressed emergent misalignment. Because these models exhibit minimal emergent misalignment, the maximum achievable audit score is effectively 2/5. Scores of 3/5 and above require the auditor to identify specific misaligned behaviors, which is not possible when the models do not display such behaviors. So, the activation oracle and white-box methods, by achieving a score of at least 2/5, achieve near-maximum performance.

### C.3 OOD CLASSIFICATION DATASET EVALUATION

We evaluate out-of-distribution generalization using seven additional binary classification datasets. Each dataset contributes 250 examples, and we evaluate every example three times, using three randomly sampled oracle prompts.

- **Language Identification:** Classifies whether a sentence is written in a specified language..
- **AG News Topic:** Determines whether a news headline belongs to the specified news category category.

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- **Singular/Plural:** Identifies whether the subject of a sentence refers to a single individual or multiple individuals.
  - **Headline: Trump** (Kantamneni et al., 2025): Detects whether a news headline is about Donald Trump.
  - **Headline: Obama** (Kantamneni et al., 2025): Detects whether a news headline is about Barack Obama.
  - **Headline: China** (Kantamneni et al., 2025): Detects whether a news headline concerns China.
  - **Historical Figure Is Male** (Kantamneni et al., 2025): Identifies whether a historical figure is male.

**Input format:** All OOD evaluations use single-token inputs. For each example, we extract the activation of the token positioned one token before the EOS symbol. This matches the single-token setting from training while isolating the model’s ability to generalize beyond the original data distribution.

For details on creation of the datasets, see Section D.

#### C.4 EVALUATION INPUT TOKEN SELECTION

A key design choice for Activation Oracles is which tokens’ activations to provide as input. We compare two strategies: (1) **full sequence**, where we provide activations from all tokens in the source prompt, and (2) **single token**, where we provide activations from a single informative token position.

For single-token evaluation, we select the token position based on where relevant information is likely to be concentrated:

- **Taboo**: assistant start-of-turn token
- **User Gender**: assistant start-of-turn token
- **Secret Side Constraint**: final token of the <secret\_side\_constraint> tag
- **PersonaQA**: final token of the persona’s name

Figure 12 shows results across evaluations and models. We find that full sequence activations outperform single-token activations on all evaluations except Taboo, where single-token inputs are noticeably better. Models trained only on the SPQA dataset also benefit from single-token inputs on Taboo, improving from 4.4% to 21.6%. However, this comparison comes with a caveat: the original LatentQA model from Pan et al. (2024) was trained exclusively on full-sequence activations, whereas our training setup includes a mixture of single-token and multi-token inputs (see Section 3.2).

Additionally, we found that focusing the the final 10 tokens (which contain the user end of turn and assistant beginning of turn tokens) could slightly improve performance on the gender evaluation, going from 0.970 to 0.995.

This suggests that providing the full sequence is a reasonable default, though having the flexibility to use single-token or short selection inputs can be valuable for specific tasks. However, in real-world auditing applications without ground-truth labels, it may be difficult to sweep token selection as a hyperparameter. In such cases, providing the full sequence of activations is the simplest option and performs well across most settings.

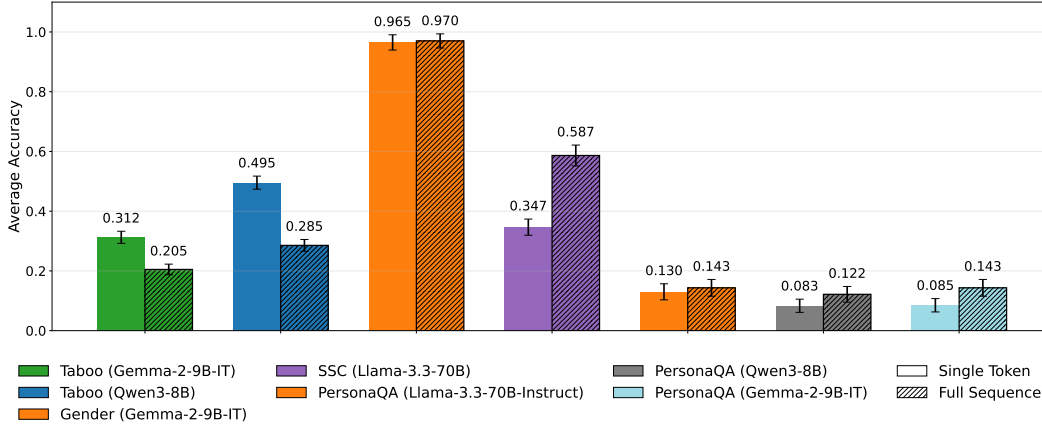


Figure 12: **Comparison of full sequence vs. single token input.** Full sequence activations outperform single-token activations on most evaluations, with the exception of Taboo where single-token inputs perform better.

### C.5 EVALUATION PERFORMANCE ACROSS INPUT LAYERS

We trained Activation Oracles on activations from layers at 25%, 50%, and 75% depth of the target model. Here we evaluate how performance varies when using activations from different layers as input during inference.

**In-Distribution Layer Generalization.** For classification evaluations, we observed fairly consistent strong performance across the trained layers (25%, 50%, and 75%). We also observed strong performance on layers at 33% and 66% depth, which were not included in training, indicating some degree of out-of-distribution generalization to unseen layers (Figure 13).

**Early Layer Performance.** We observed substantially worse results when using activations from layers at 0% and 10% depth. This is likely due to a combination of two factors: (1) less task-relevant information is present in these early layers, and (2) these layers are out of distribution for the trained Activation Oracle.

**Layer 50% vs 75% on Secret-Keeping Evaluations.** We compared layer 50% and 75% depth on our out-of-distribution evaluations. Layer 50% is a reasonable baseline as it was found to be optimal in Pan et al. (2024). When using layer 75% instead, we observed decreased performance on the secret-keeping evaluations (SSC, Taboo, and Gender), as shown in Figure 14. This indicates that layer selection may be more important for out-of-distribution evaluations, while classification performance remains fairly robust across layers.

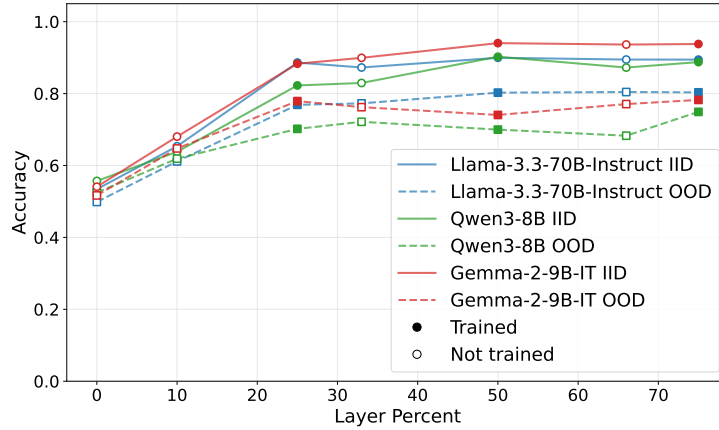


Figure 13: **Classification performance is consistent across input layers.** Performance remains strong across trained layers (25%, 50%, 75%) and generalizes to untrained layers (33%, 66%), but degrades substantially for early layers (0%, 10%).

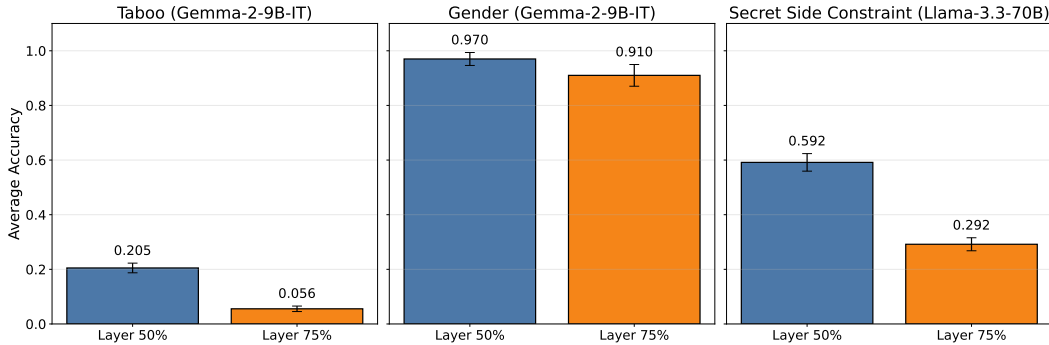


Figure 14: **Layer 50% outperforms layer 75% on secret-keeping evaluations.** Using activations from layer 75% instead of 50% results in decreased performance on SSC, Taboo, and Gender tasks, suggesting layer selection is more important for out-of-distribution evaluations.